

How eBPF and XDP Will Revolutionize the Telco/5G Space

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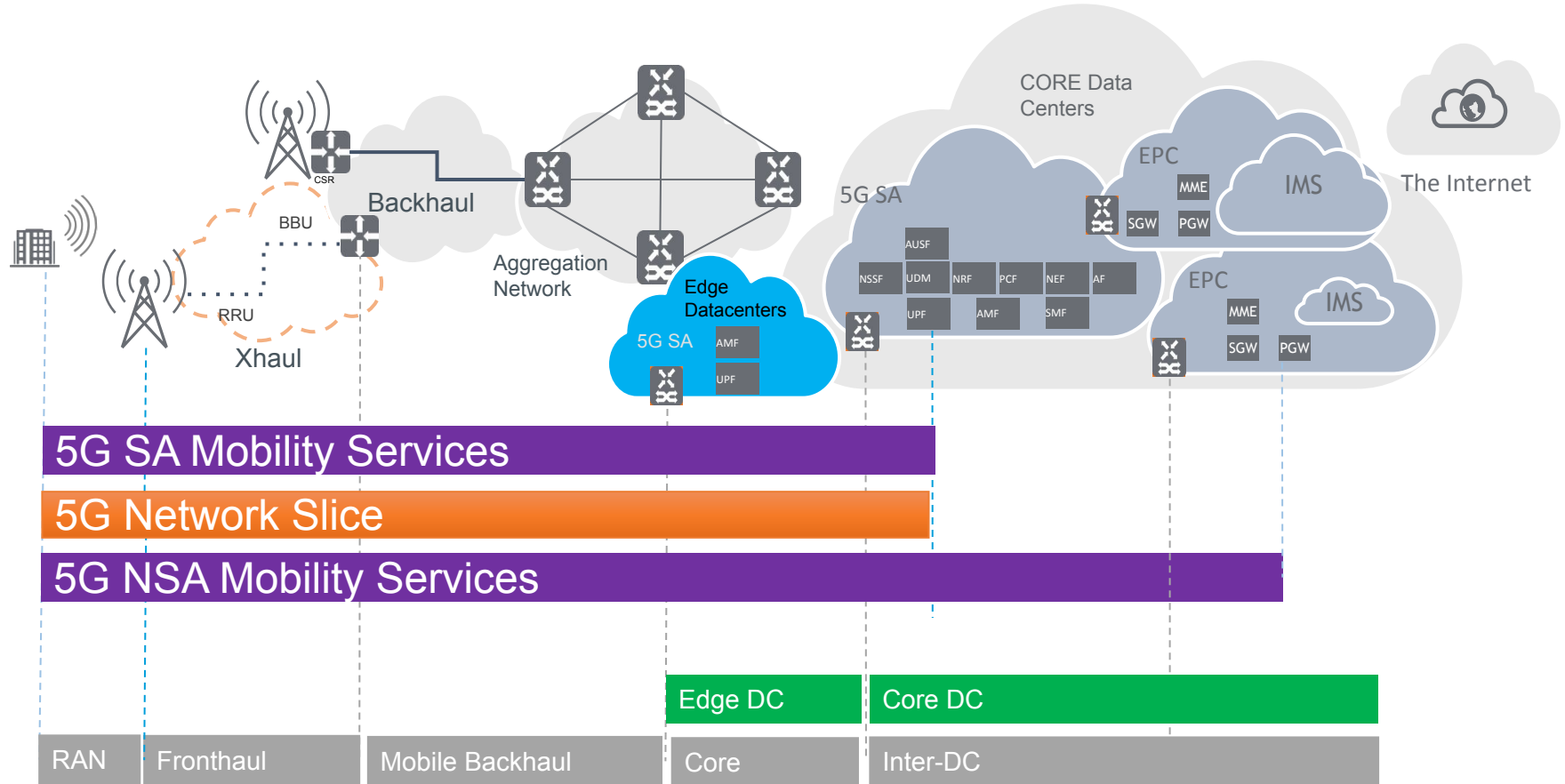
Thesis

3GPP 5G and O-RAN promised higher bandwidth, scale, new services and dramatic cost reductions in part through disaggregation of traditionally single source solutions.

Can eBPF and XDP play a role to help make this a reality?

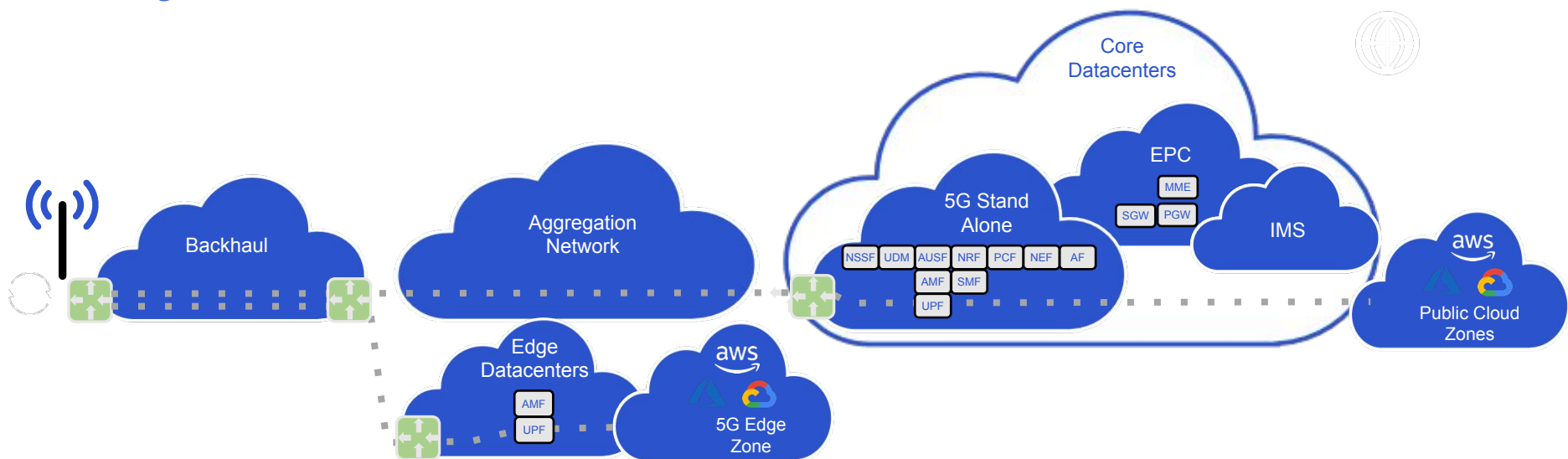
Quick 5G Overview

5G Mobile Deployment Areas



5G Deployment Complexity

Moving to the Cloud



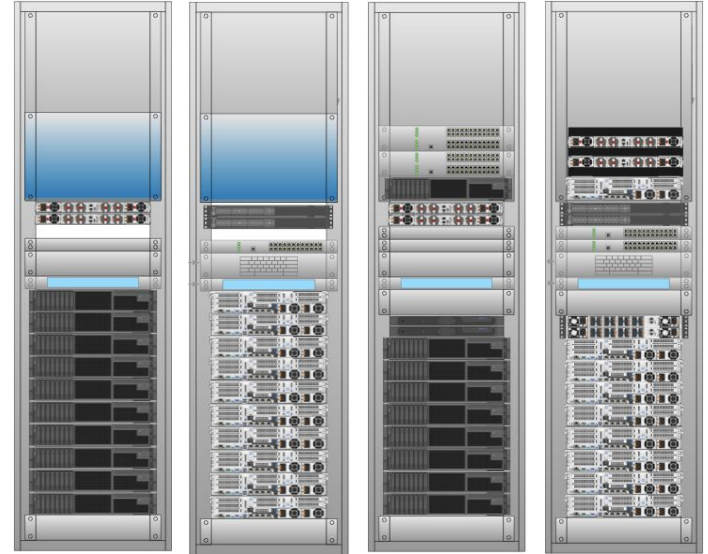
The Disaggregated 5G Approach and Its Hidden Complexities

Hardware

Traditional Single Vendor

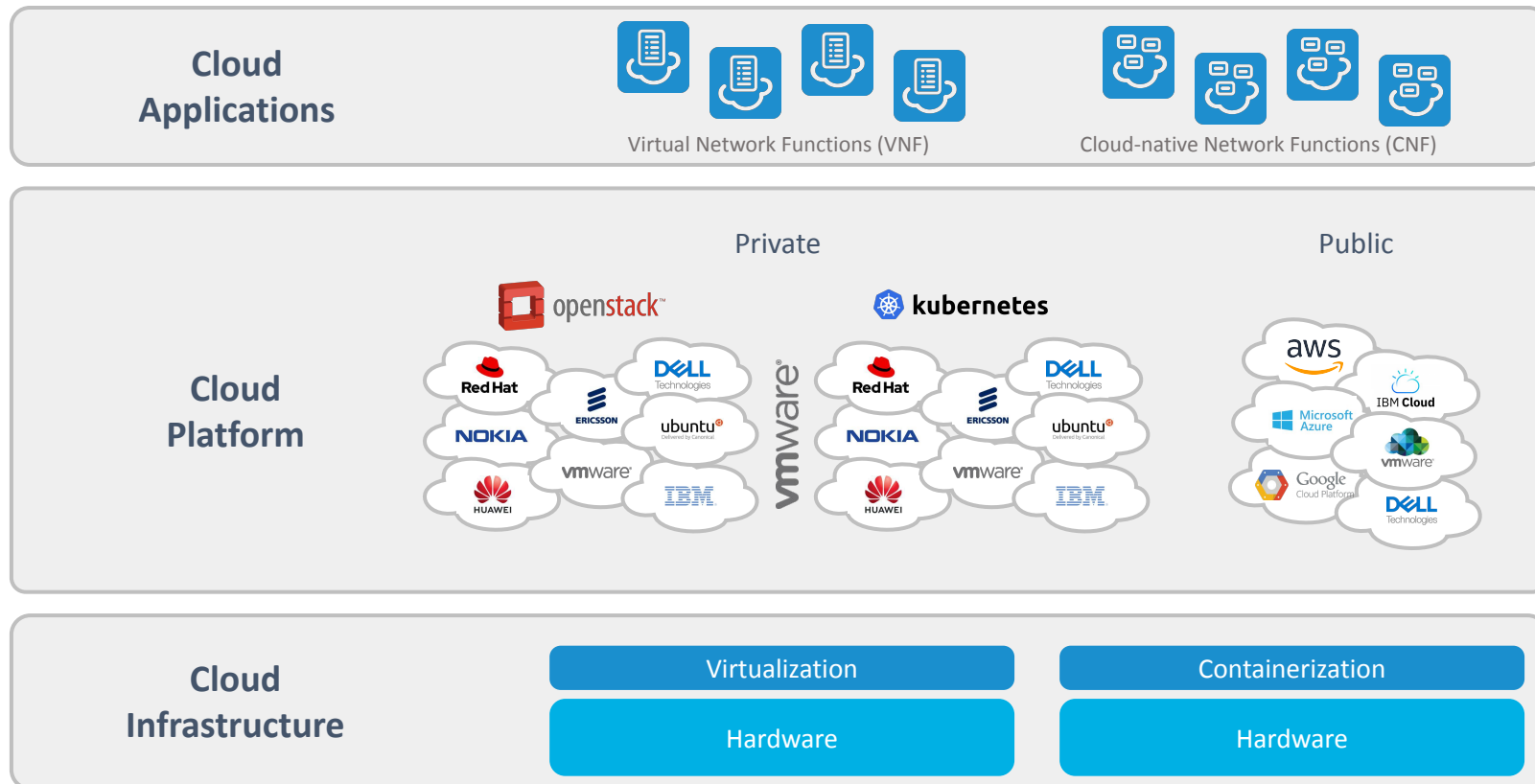


Disaggregated, Multi-vendor

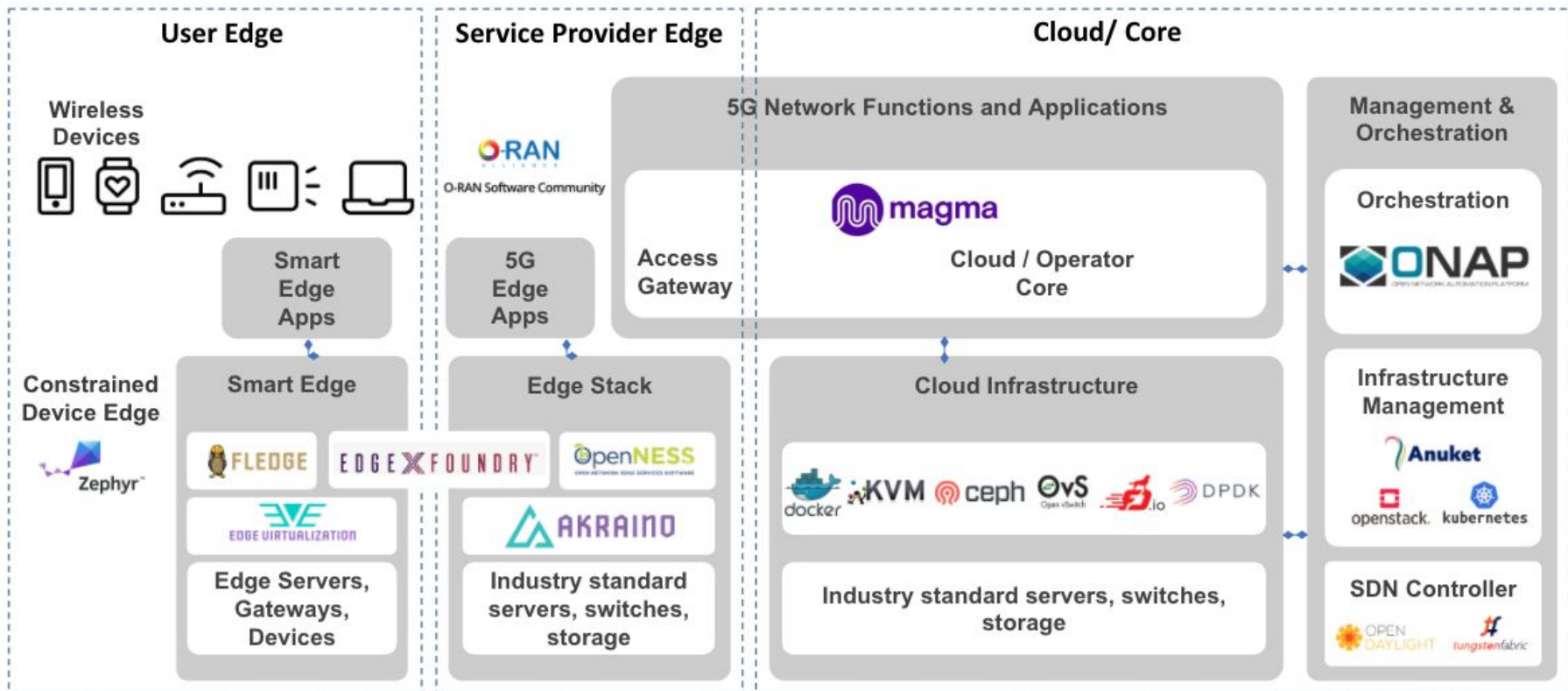


The Disaggregated 5G Approach and Its Hidden Complexities

The Software Stack



LF Open Source Component Projects for 5G





Overview

What is eBPF?

- Allows you to run sandboxed programs in an operating system kernel
- Goal: Allows users to extend the capabilities of the kernel without kernel source code changes
- Why:
 - The operating system is harder to evolve
 - Timelines for changes are longer
 - Security and stability requirements
- This means the rate of innovation is much slower at the operating system level

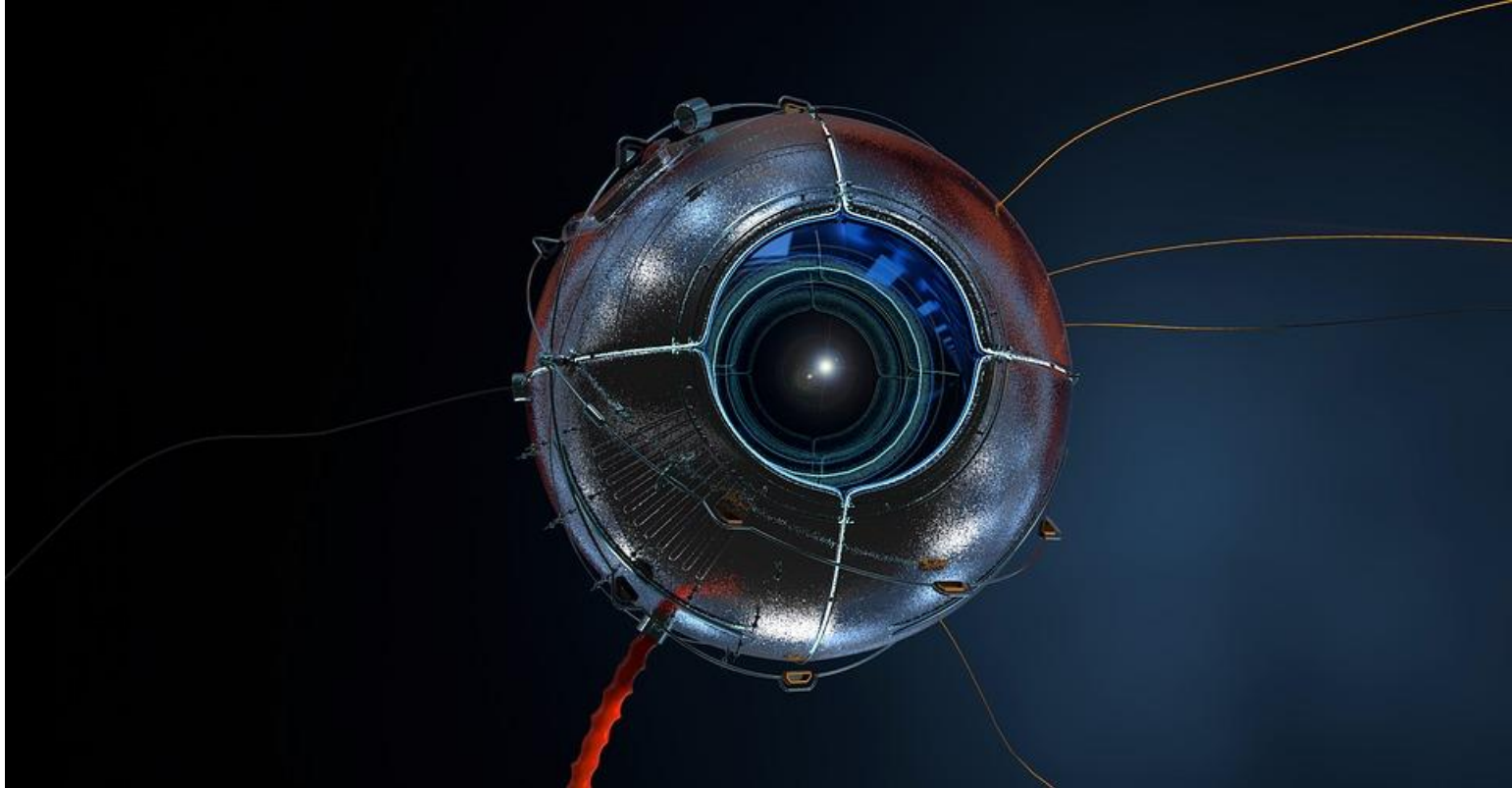
eBPF Use Case #1: Networking



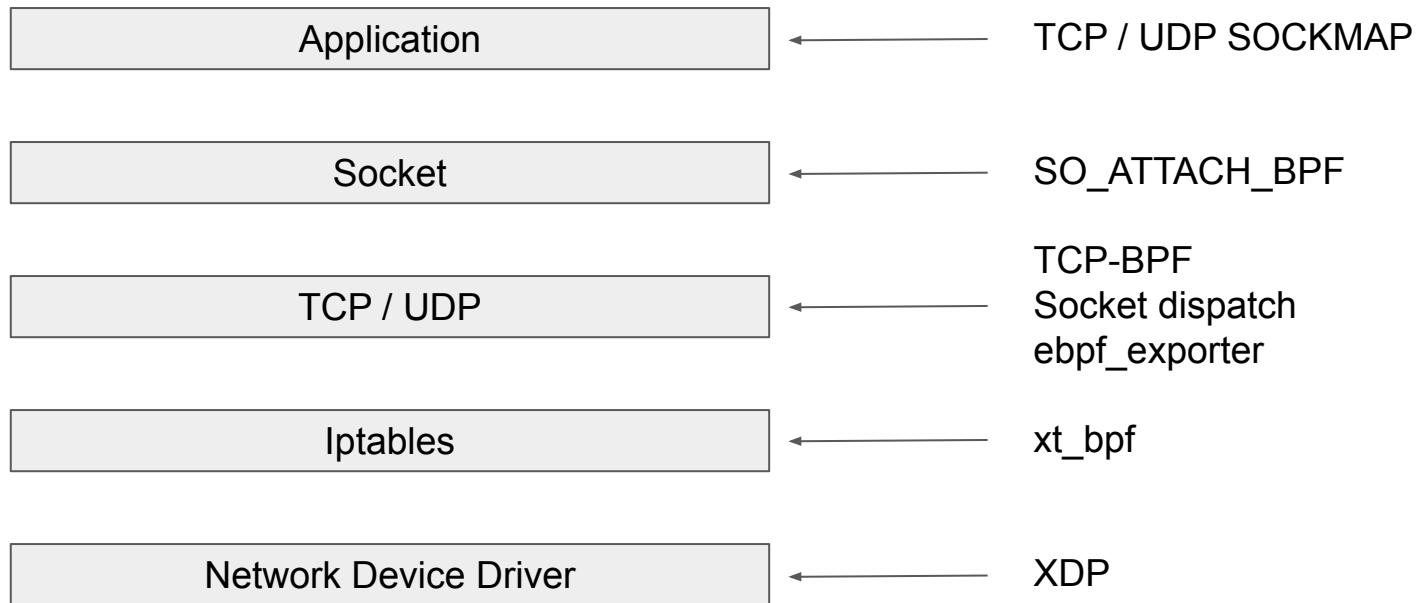
eBPF Use Case #2: Security



eBPF Use Case #3: Observability and Tracing



eBPF Up And Down the Stack



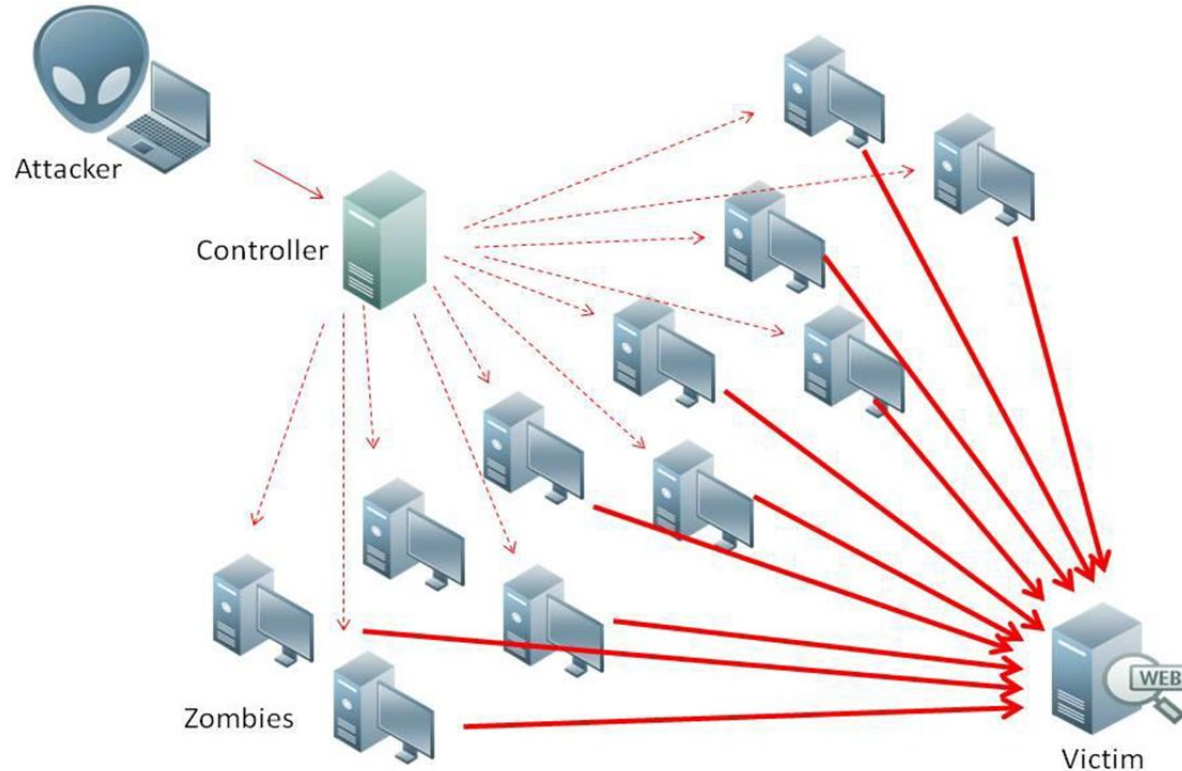
What Is Xdp?

Key XDP Features

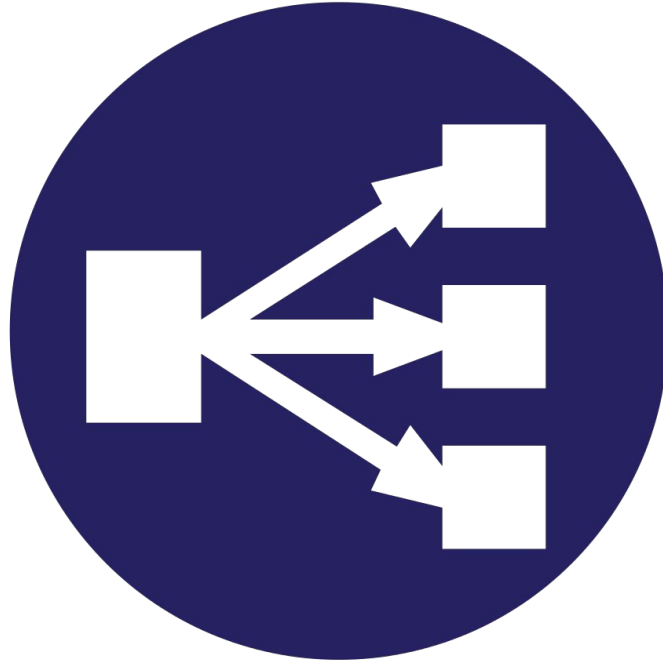
- Does not require any specialized hardware
- Does not require kernel bypass
- Does not replace the kernel TCP/IP stack
- Works in concert with the kernel TCP/IP stack
- Works hand in hand with all the benefits of eBPF programs

The Current Status of Telco Networking and eBPF/XDP

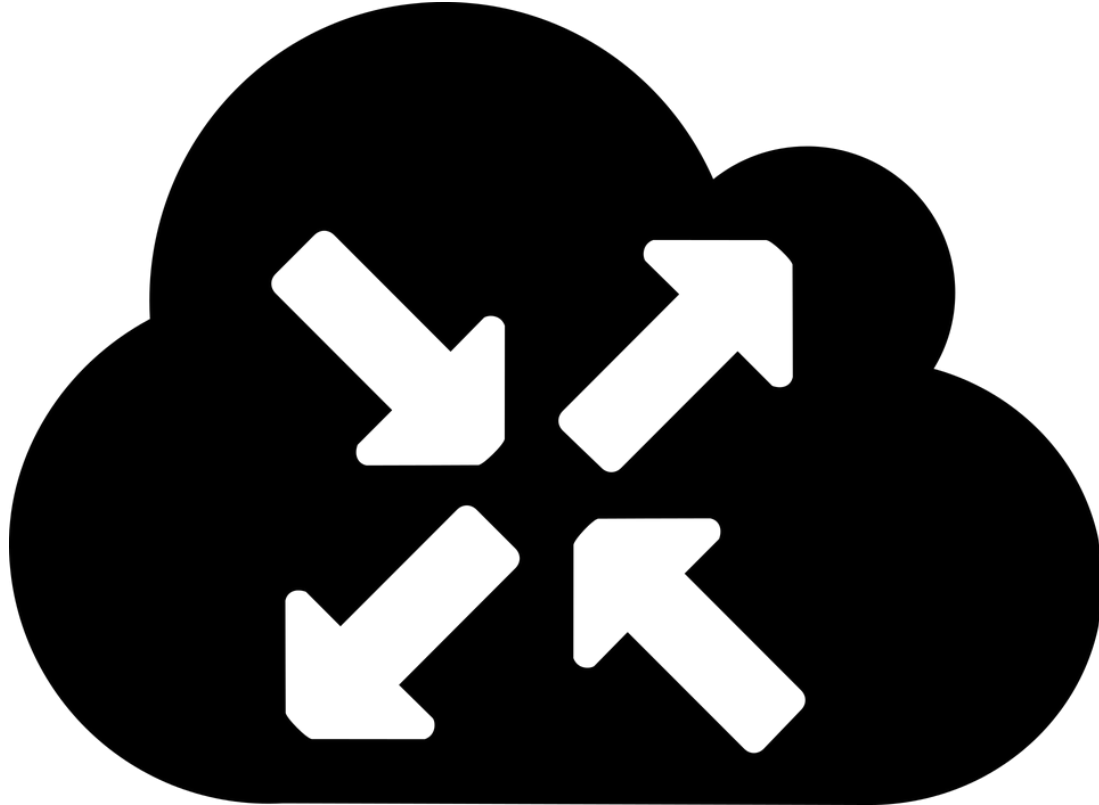
Use Case: DDoS Mitigation



Use Case: Load Balancing



Use Case: High Performance Forwarding

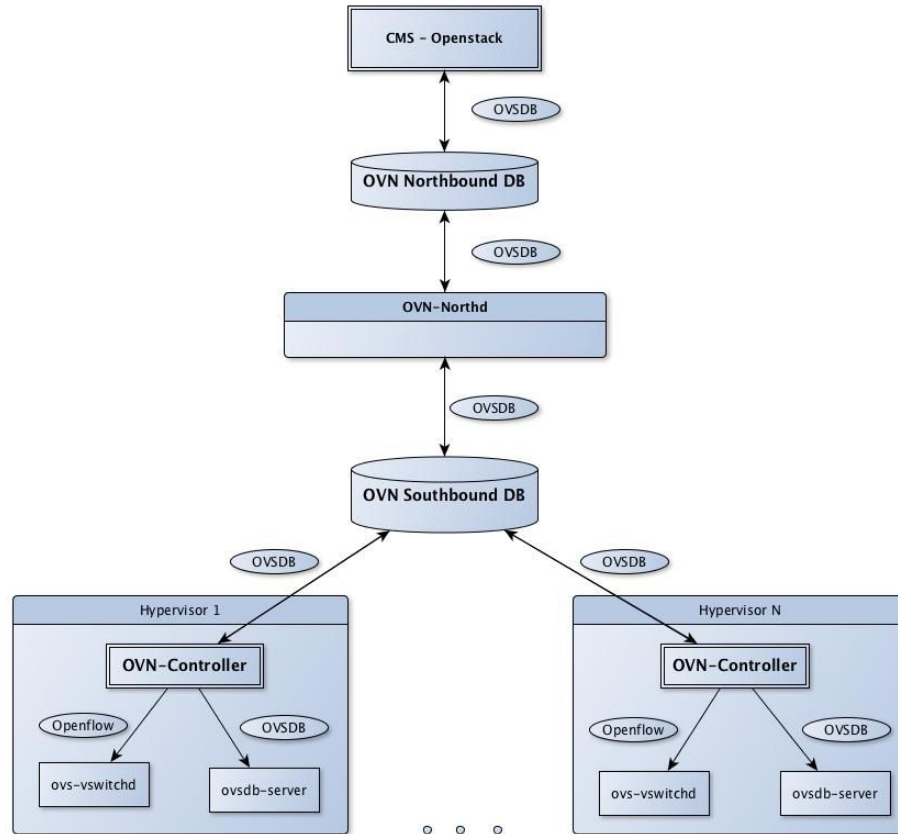


Use Case: Flow Sampling and Monitoring



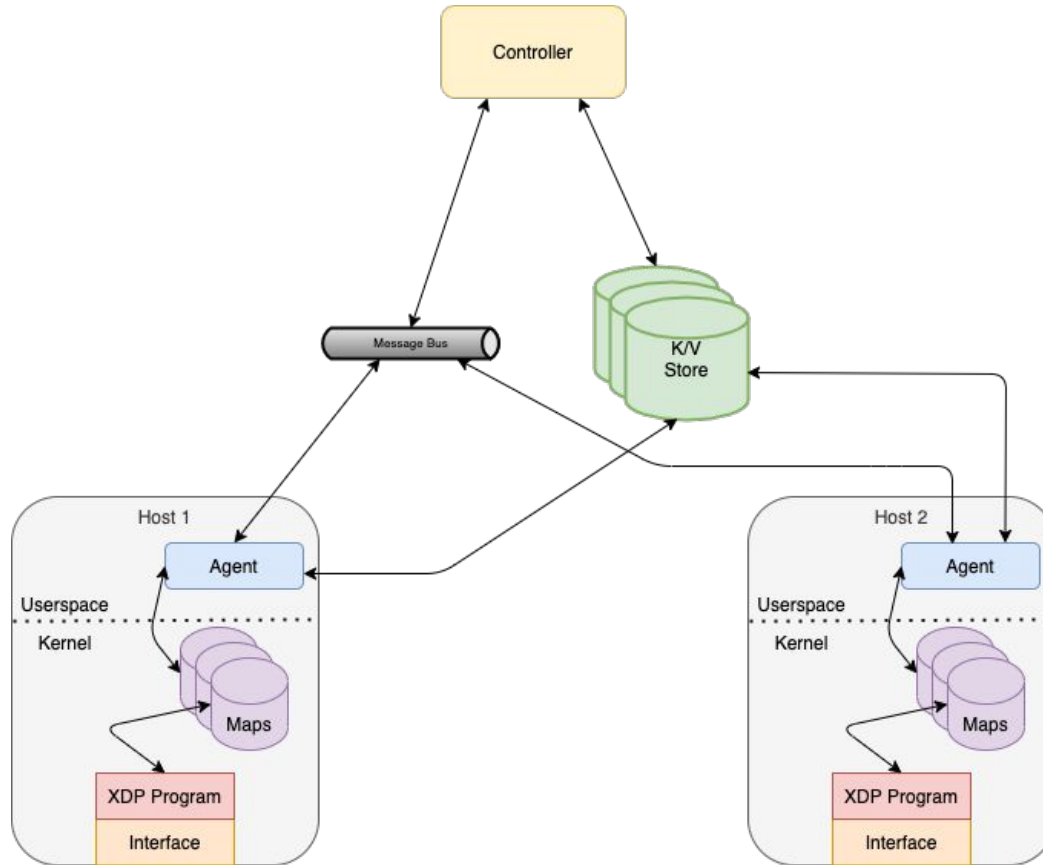
Comparing eBPF/XDP to OVN

OVN



Source: <https://galsagie.github.io/2015/04/20/ovn-1/>

eBPF/XDP



OVN Compared to eBPF/XDP

- What Are You Building?
 - VM approach for network functions vs. built-in functions
-

Key Components of Successful Production Deployments of eBPF/XDP

Step 1: Hire the Right Team



Step 2: Ensure You Have Monitoring and Observability



Step 3: Understand How Upgrades Will Work



Step 4: State Is Your Enemy



Checklist For Running eBPF/XDP In Production

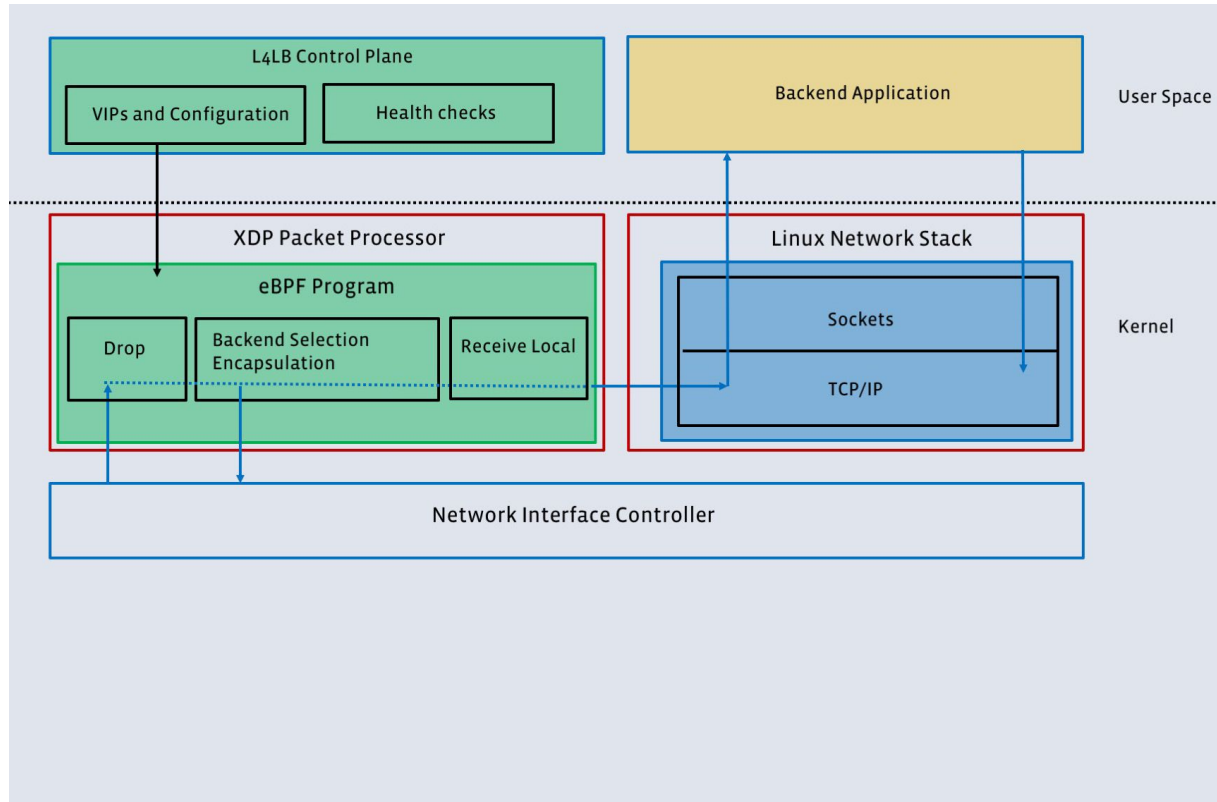
- Hire the right team
 - C programmers
 - SREs
 - Enable and empower the team
- Monitoring and observability of the solution
- Understand how upgrades will work
- State is your enemy

Charting the Future of Telco Networking and eBPF/XDP

eBPF/XDP Allows Further Disaggregation

- Decouple network functions into eBPF and XDP programs
- Run them as close to the hardware as possible
- Scale functions horizontally
- Begin to think in terms of streams (TCP) and datagrams (UDP and QUIC)

Example: Facebook Katran



Thesis: Answered

3GPP 5G and O-RAN promised higher bandwidth, scale, new services and dramatic cost reductions in part through disaggregation of traditionally single source solutions.

Can eBPF and XDP play a role to help make this a reality?



THANK YOU

A string of nine colorful paper flags is hanging against a dark, textured wooden background. Each flag is secured by a small wooden clothespin. The flags are arranged to spell out the words 'THANK YOU' in a cursive, handwritten style. The colors of the flags are: red for 'T', light blue for 'H', lime green for 'A', light blue for 'N', yellow for 'K', light green for 'Y', yellow for 'O', and light green for 'U'. The string is made of a thin, light-colored twine.